

Exploring Empathy with Digital Humans

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ABSTRACT

Digital humans are autonomously-animated virtual people whose social interactions are driven by artificial intelligence. They are increasingly being deployed in applications such as healthcare, customer service, and education, and they may have a place in the metaverse. For digital humans to have effective social relationships with users, it is important that they are capable of empathetic interactions. This research aims to evaluate and build upon the autonomous empathy system of a digital human through five experimental studies. Psychological and physiological data will be collected, and the effects will be compared in an Augmented Reality environment and cross-culturally. This paper presents the research agenda, and discusses considerations and challenges for empathetic interactions with digital humans.

Keywords: Empathy, digital humans, autonomous animation, human-computer interaction, emotion, affective behavior, social agents

Index Terms: J.4 [Social and Behavioral Sciences]; H.5.2 [Information Interfaces and Presentation (e.g., HCI)]; User Interfaces; C.3 [Special-Purpose and Application-Based Systems]

1 INTRODUCTION

Digital humans are a new form of embodied conversational agent with hyperrealistic, humanlike embodiments that use artificial intelligence to inform interactions with people [1]. They can use live neural networks to make classifications of peoples emotional states during interactions and respond using a combination of appropriate speech, facial behaviors, and body gestures. Digital humans are increasingly being deployed in applications including healthcare, education, and customer service, and in future they may have a role in the metaverse. For digital humans to be effective and engaging in these roles, it is important that they can autonomously engage with people in an empathetic manner.

According to the Perception Action Model, empathy refers to the ability to perceive and share the emotions of others, and it can be split into affective, cognitive, and somatic components [2]. *Affective empathy* refers to the emotions people feel in response to another person's emotions, whereas *cognitive empathy* refers to the ability to identify the emotions that others are feeling. *Somatic empathy* describes the physiological responses people experience towards others' emotions. Empathy has been associated with increased activity in particular brain regions (e.g., the frontal lobe and central parietal regions), and the activation of mirror neurons [3] [4]. Moreover, empathy has been shown to affect the peripheral nervous system (e.g., skin conductance, heart rate) and motor responses (e.g., facial EMG, startle blinks) [5] [6].

Empathy is important in social relationships because it allows people to understand the emotional states of others and respond in appropriate ways. When delivered appropriately, empathy has been shown to reduce emotional and physiological distress in the

receiver [7], and improve relationship quality [8]. Hence, it is an important social skill for digital humans to master. Indeed, research has shown that users value empathy in interactions with digital humans [9], and with embodied conversational agents more broadly [10]. However, further research is needed to understand how to best model empathy in digital humans.

1.1 Empathy from digital humans

Embodied agents and social robots have only recently started to be programmed with capabilities to detect and respond to user emotions. Approaches to modeling empathy in these technologies have varied in complexity and in terms of whether multimodal data is incorporated in the model [11] [12] (e.g., speech and gesture recognition), or whether singular data types are focused on (e.g., facial expressions, speech prosody) [13] [14]. Recent advances in machine learning, data availability and computing power have given rise to autonomous, multimodal emotion detection systems that could inform more accurate empathy responses and be used by digital humans. One study found that an automated empathy system in a virtual game character was rated as equally appropriate and accurate as when the expressions were human controlled [15]. However, further research is needed to explore the effect of automated, multimodal empathy systems in digital humans. It remains unclear what techniques result in the most effective digital human empathy response model, and whether digital human empathy can improve psychological and physiological outcomes for people in a manner akin to human empathy. Albeit, promising results have been found with social robots and embodied agents, whose empathy has been shown to improve psychological and physiological distress for users in some studies [16] [17].

1.2 Empathy towards digital humans

Research has shown that people can experience cognitive and affective empathy towards embodied agents, virtual characters, and robots [18] [19]. In experimental studies, people have shown cognitive empathy towards robots by attributing more negative feelings towards them when they are being tortured compared to when they are treated nicely [20]. Similar results have been found in embodied agents [18]. People have also reported feeling affective empathy towards robots, as evidenced by reporting less positive and more negative feelings after watching a robot in pain [20].

These findings show that people are capable of subjective, cognitive and affective empathy responses towards social robots and embodied agents appearing to be in emotional distress. Although, most of the literature focuses on agents with predetermined, preprogrammed emotional expressions; relatively few studies have looked at empathy towards agents that are autonomously generating their own emotions in real time.

There is a paucity of literature evidencing objective, somatic empathy responses towards embodied agents, such as changes in neurological, physiological, and motor responses. However, one study in robots found that people experienced significantly

increased skin conductance while watching a robot experience torture, compared to the robot being treated nicely [20]. Although, no significant effects were found for heart rate. Further research is needed to investigate whether people can feel somatic empathy towards embodied agents, including digital humans, using a broader array of physiological markers.

Moreover, alternative delivery environments such as Augmented Reality (AR) and Virtual Reality (VR) may amplify empathetic responses towards digital humans because of their increased social presence [21]. Indeed, research to date has shown that storytelling in VR improves perspective taking compared to other delivery formats [22]. AR and VR delivery formats present new opportunities for empathetic interactions with digital humans.

1.3 Present research

This paper presents a research agenda for five experimental studies that aim to understand the experience of empathy with a digital human by investigating the following research questions:

1. Do people perceive the digital human as empathetic towards others?
2. Do people perceive the digital human as empathetic towards themselves?
3. Do people feel empathy towards the digital human?

It is hypothesized that a digital human who displays verbal and non-verbal empathy cues triggered by an automated system will be perceived as more empathetic towards other people and the user, and the user will experience more empathy towards the digital human compared to a digital human displaying neutral cues. The results will be used to inform improvements to the digital human's autonomous empathy system at each phase of the research.

2 METHODS

A multiphase research project will be conducted, with each phase building on the results of the previous. Five experimental studies will be run with a parallel group design to compare the performance of the automated empathy system to a neutral control in an emotional storytelling scenario. In each study, participants will be randomly allocated to either the empathetic or neutral digital human condition (1:1 allocation ratio). Each experiment will recruit 88 participants (44 per group), based on the results of a power analysis with G*Power software (with parameters of a fixed effects omnibus one-way ANOVA with 2 groups, 80% power, $F=0.4$ [15], and an alpha error probability of 0.005 as correction for multiple comparisons). The experiments will include a video study, a conversational study, a physiological study, an augmented reality study, and a culturally-focused study, as outlined in greater detail in the following sections. The video study aims to provide a preliminary evaluation of the empathy system's current performance compared to a neutral control. This will be followed by a conversational study that incorporates feedback from the previous study and evaluates an updated system. The conversational study will evaluate performance during direct interaction with many users, compared to the video study in which participants watch the digital human interact with a confederate. Next, a physiological study will replicate the conversational study design but in addition collect physiological data to assess somatic empathy in response to an updated empathy system. An augmented reality study will assess whether the effects of the digital human empathy responses are amplified in an immersive interaction environment. The final phase of the research involves culturally tailoring the empathy system to evaluate how this affects system performance.

Participants will be adults aged 16 years or older with English fluency. For studies involving a live interaction with the digital

human (i.e., all apart from the video study), participants will need to enjoy acting to take part. Recruiting people who enjoy acting increases the chance that they will be willing and able to act the emotional scenarios and trigger the empathy system.

2.1 Video study

An online video study will be conducted with adults aged 16 years or older with fluent English and, if recruited from Amazon Mechanical Turk, a 'Master Worker' status (i.e., a registered worker who has reliably shown to have a high-quality output). Participants will be recruited from advertisements posted to Amazon Mechanical Turk and the University of Auckland website. The study will be hosted on Qualtrics, a secure survey website.

After providing written informed consent, participants will complete a baseline questionnaire on demographics (e.g., age, gender, ethnicity, education level, occupation, country of residence) and trait empathy (Interpersonal Reactivity Index [23]). Next, participants watch ten minutes worth of videos of a digital human talking with an actor. In the first five minutes of the video, the actor will perform different emotional scenarios as the digital human listens and responds. The emotional scenarios are adapted from previous research [24]. While listening, the digital human will use neural network models to identify the actor's emotions and then either generate empathetic or neutral responses, depending on the condition to which the participant has been randomly allocated.

After watching each emotional scenario, participants will rate how empathetic the digital human was towards the actor using a 100mm visual analogue scale [25]. Participants will rate the appropriateness of the digital human's responses using five items with a 5-point likert scale and response options ("strongly disagree" – "strongly agree"), taken from prior research [15].

Next, participants will watch another 5 minutes of video in which the digital human tells its own emotional stories to the actor while the actor responds. The participant will view a digital human with emotional expression or a neutral digital human, consistent with the condition to which they had been randomly allocated to for the first video. Participants will be asked to describe what emotion the digital human was feeling in a one-word response, in order to assess *cognitive empathy*. Participants will also report which emotion they feel after watching the digital human tell her story in a one-word response, to assess *affective empathy*. Two 100mm visual analogue scales will be used to assess how empathetic the participant felt towards the digital human [26], and how well they related to the digital human's story using response anchors (0 = "not at all" – 100 = "completely"). One item will assess whether the digital human's facial expressions and gestures matched what she said using a 5-point likert scale with response anchors (0 = "strongly disagree" – 4 = "strongly agree"). At present, it is anticipated that measures will be administered after each of the 10 emotional scenarios are performed, therefore brief measures were chosen to reduce participant burden. Should the study design shift to assess measures at the end of a monologue comprised of many scenarios (as opposed to after each scenario), more detailed assessments will be sought for cognitive and affective empathy.

2.2 Conversational study

A conversational study will be conducted with similar methodology to the video study, except that participants will directly interact with the digital human instead of watching videos online. Participants will take the place of the actor in the video study and work through the same emotional scenarios with a digital human.

Participants will be adults aged 16 years or older with fluent English who enjoy acting. Recruitment will take place through flyers posted at the University of Auckland and on Facebook groups for community theatre and drama classes.

The experiment will take place in a private research room at the University of Auckland, New Zealand. Audiovisual data will be collected, and five research assistants will view the recordings to rate the ability of each participant to act the scenarios in an emotive way using a standardized form. This will enable evaluation of how well the digital human responds to participants who are highly emotionally expressive compared to less emotionally expressive.

2.3 Physiological study

A physiological study will be conducted to assess *somatic empathy* responses. The physiological study will be conducted in a private room at the Empathic Computing laboratory at the University of Auckland. The study will use similar methodology to the conversational study, however additional data will be collected on physiological indicators of empathy including heart rate and skin conductance (via Empatica E4), brain electrical activity (via EEG), and facial muscle activity (via facial EMG).

2.4 Augmented reality study

An AR study will be run that is identical to the physiological study except that participants will interact with the digital human in an AR environment using an AR headset. Facial EMG data will be collected if it is possible with the AR headset. An AR study will be conducted as it is possible that people may experience heightened empathy with a digital human in AR due to increased feelings of social presence. Data will be collected in a controlled, private room that is suitable for using a HoloLens2 AR headset at the Empathic Computing laboratory at the University of Auckland. Participants will be provided with a brief training and orientation to the AR headset prior to the interaction.

2.5 Cultural study

The final phase of the research involves a cultural study in which the emotional behaviour of the digital human is varied based on the cultural norms of the countries that the research is conducted in. At this stage, it is anticipated that the countries will be the United States, Japan, a European country, and New Zealand. In each country, there will be a multitude of ethnic groups and we will carefully apply cultural models that are developed in consultation with experts from the cultures involved. Although the earlier phases of the research collect participant ethnicity data as a variable to be considered in the analyses, this phase specifically explores culture as the main variable of interest. A thorough and prolonged consultation process will take place with cultural experts prior to the research design and model adaptation processes. The methodology will be refined through consultation with cultural experts, however it is anticipated that this study might adopt a similar methodology to the conversational study. It is anticipated that the study will be conducted online, however participants located in Auckland may take part in person from a private room at the University of Auckland.

2.6 Digital human

The digital human that will be used in the research is “Sam” by Soul Machines (Auckland, New Zealand) (see Figure 1). Sam is driven by an autonomous neurobehavioural computing framework consisting of artificial neural networks that inform social and emotional behaviours [1]. As part of her empathy system, Sam

classifies the emotional states of users in real time based on their facial and vocal data. Sam uses this information to autonomously provide an empathetic response, delivered through verbal and non-verbal communication cues (e.g., facial and body gestures). Sam contains an internal affective system that is impacted by the emotional states she detects in users. In the empathetic condition, Sam’s empathy system is fully activated. In the neutral condition, Sam will make classifications of users emotions however she will be programmed to not let these classifications affect her responses. Rather, she will maintain a neutral facial expression throughout the interactions.



Fig. 1. “Sam”, a digital human by Soul Machines (Auckland, New Zealand).

2.7 Data analysis

For each study, multiple ANOVAs with Bonferroni correction will be conducted to investigate the effect of the digital human empathy system versus a neutral control on outcomes across multiple emotional scenarios.

3 PROGRESS TO DATE

Institutional review board (IRB) approval has been sought for studies 1 to 4. Protocols for each experiment will be published on OSF Registries prior to data collection. Data collection will commence in early 2022 following ethics approval. Consultation processes for the cultural study will begin in 2022 to inform the research approach, design, and digital human behaviours. Once these aspects have been decided, IRB approval will be sought for the study.

4 DISCUSSION

This paper presents a research agenda for evaluating and building upon an automated empathy system in digital humans. Considerations for developing automated empathy systems, and strengths and limitations of the proposed research agenda will now be discussed.

Researchers looking to create autonomous empathy systems in digital humans should bear in mind several factors that could influence results. Firstly, similarity between the digital human and the user may affect empathetic responding. Research has shown that people tend to feel greater empathy towards others who are similar to themselves (e.g., demographically, psychologically) [6]. Thus, we might expect users to feel more empathetic towards digital humans who are characteristically similar to them. Secondly, differences in cultural norms for emotional expression

could affect how users judge the digital human's emotional responses [27]. For example, east Asian cultures have been shown to prefer expressing low arousal positive affect (e.g., calm), and to avoid overtly expressing negative emotions with a tendency to report somatic symptoms instead [27]. Whereas in North American white culture, high arousal positive affect is preferred (e.g., excitement) and negative emotions are overtly expressed. This may affect how digital humans should interpret emotional cues from people of different cultures as part of their empathetic response, as well as indicate possible variation in how people may interpret the digital human's emotional expressions. Digital humans that violate the user's cultural norms may be less effective with their empathy. Thirdly, the user's level of trait empathy may affect how they perceive and respond to the digital human, independently of the digital human's behaviour. High trait empathy has been associated with greater empathetic responses towards robots and embodied agents, irrespective of their design [20].

A strength of this multiphase research project is that it consists of several experiments that collect subjective and objective data to better understand experiences of empathy with a digital human and to inform updates to an empathy model. The combination of subjective and objective data helps form a robust evaluation, and enables assessment of different types of empathy responses (e.g., cognitive, affective, somatic). Moreover, the empathy model performance will be tested in alternative delivery environments and cultural contexts.

Limitations of the research include that the results could be affected by demographic matching between the digital human and the user, and by cultural norms for emotional expression before cultural tailoring is introduced in experiment five. Due to practical constraints, it is not feasible to match the digital human to each user demographically. It is also possible that unanticipated discomfort during the research (e.g., dizziness from the augmented reality headset) could interfere with the data, although steps will be taken to prevent this (e.g., exclusion of individuals susceptible to motion sickness, AR headset training).

5 CONCLUSION

Digital humans that are capable of autonomously providing empathetic responses to people will be more effective and engaging in their roles. Moreover, digital humans that are capable of evoking empathy in users may form closer relationships with them. This paper proposes a research programme to evaluate the effects of the autonomous empathy system of a digital human on user psychology and physiology. Effects will be compared in an augmented reality environment and cross-culturally. The results will provide a framework for evaluating and building upon a multimodal, autonomous empathy system for use in digital humans that may be used in a variety of applications.

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